Probability and Statistics

Saydric Zheng

July 26, 2026

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1 Univariate Discrete Probability Distribution

Random Variable: X is a function ($X : S \to \mathbb{R}$) that assigns a real number to each point in a sample space S.

Discrete Random Variable: a set is countable if its element placed in one-to-one.

Probability Mass Function: $f(x) = \mathbb{P}(X = x)$ for all $x \in \mathbb{R}$. The set of pair $\{(x, f(x)) : x \in \mathbb{R}\}$ is the probability distribution

PMF properties:

(1) $f(x) \geq 0$ for all $x \in \mathbb{R}$

(2) $\sum_x f(x) = 1$

Cumulative Distribution Function(CDF): $F(x) = \mathbb{P}(X \leq x)$
CDF properties:

(1) $0 \leq F(X) \leq 1$ for all $x \in \mathbb{R}$.

(2) $F(X)$ is non decreasing function for all $X \in \mathbb{R}$.

(3) $\lim_{x \to -\infty} F(X + h) = 0$.

(4) $\lim_{h \to \infty} F(X + h) = 1$

(5) $\lim_{h \to 0^+} F(X + h) = F(X)$.

Remark: $f(a_i) = F(a_i) - F(a_{i-1})$ for $x \geq 2$

Function of Random Variable: If X is a discrete random variable with range $\mathbb{A}$, then $Y = g(X)$ is also discrete random variabel, and its pmf can be determined using the pmf of X.

$$f_Y(y) = \mathbb{P}(Y = y) \quad \text{for any value of } y$$
$$g(x) = y, \{x \in \mathbb{A} : g(x) = y\}, \quad \text{define} : f_X(x) = \mathbb{P}(X = x)$$
$$f_Y(y) = \sum_{x \in \mathbb{A} : g(x) = y} f_X(x)$$

Exmaple: X represent the sum of upturned faces o the two thrown dice. Determine the probability distribution
$Y = x \bmod 5$.

$$f_Y(y) = \mathbb{P}(Y = y) = \mathbb{P}(X = g(y)^{-1}) = f_X(g(y)^{-1})$$

1.1 Expetation of Random Varibale

Expected Value:

$$\mathbb{E}(x) = \sum_{\text{all } x} x f(x) = \sum_{x \in \mathbb{A}} x f(x)$$

Remark: expected value might can't gained.

Theorem: Suppose X os discrete random variable with range A and $pmf f(x)$. Then the expected value of some real valued function $g(x)$.

$$\mathbb{E}(g(X)) = \sum_{x \in \mathbb{A}} g(x) F(x)$$

Proof.

$$\mathbb{E}(g(X)) = \mathbb{E}(Y) = \sum_{\text{all } y} y f_Y(y)$$

where $f_Y(y) = \sum_{x \in \mathbb{A}, g(x)=y} f(x)$ Let $D_y = \{x \in \mathbb{A} : g(x) = y\}$, $f_Y(y) = \sum_{x \in D_y} f(x)$

$$\begin{aligned}
\mathbb{E}(Y) &= \sum_{\text{all } y} \sum_{x \in D_y} y f(x) \\
&= \sum_{\text{all } y} \sum_{x \in D_y} g(x) f(x) \\
&= \sum_{x \in \mathbb{A}} g(x) f(x)
\end{aligned}$$

□

Theorem: For real constant $c_1, c_2, ..., c_n$ and real valued function $g_1, g_2, ..., g_n$.

$$\mathbb{E}(\sum_{i=1}^{n} c_i g_i(x)) = \sum_{i=1}^{n} c_i \mathbb{E}(g_i(x))$$

Proof.

$$h(x) = \sum_{i=1}^{n} c_i g_i(x)$$

Substitute into the expected value formula.

$$\begin{aligned}
\mathbb{E}(h(x)) &= \sum_{x \in \mathbb{A}} h(x) f(x) \\
&= \sum_{x \in \mathbb{A}} (\sum_{i=1}^{n} c_i g_i(x)) f(x) \\
&= \sum_{i=1}^{n} c_i (\sum_{x \in \mathbb{A}} g_i(x) f(x)) \\
&= \sum_{i=1}^{n} c_i \mathbb{E}(g_i(x))
\end{aligned}$$

□

Corollary: For real constant a and b.

$$\mathbb{E}(aX+b) = a\mathbb{E}(X) + b$$

Proof. Let $c_1 = a, c_2 = b, g_1(x) = x, \textbf{and}, g_2(x) = 1$, then by theorem $\mathbb{E}(cg(X)) = c\mathbb{E}(X)$

$$\begin{aligned}\mathbb{E}(aX+b) &= a\mathbb{E}(X) + b\mathbb{E}(1)\\ &= a\mathbb{E}(X) + b\sum_{\text{all } x\in\mathbb{A}} 1\cdot f(x)\\ &= a\mathbb{E}(X) + b\end{aligned}$$

□

Remark: If $g(x)$ is not linear function , then $\mathbb{E}(g(x)) \neq g(\mathbb{E}(X))$
Variance: The average square distance from variable x to the mean μ.

$$\text{Var}(X) = \mathbb{E}((x-\mu)^2) \Rightarrow \mathbb{E}((x-\mu)^2) = \sum_{x\in\mathbb{A}}(x-\mu)^2 f(x)$$

Standard Deviation:

$$sigma = SD(x) = \sqrt{\text{Var}(x)} = \sqrt{\mathbb{E}(X-\mu)^2}$$

Theorem:

$$\text{Var}(X) = \mathbb{E}(X^2) - \mu^2$$
$$\text{Var}(X) = \mathbb{E}(X(X-1)) + \mathbb{E}(X) - \mu^2$$

Proof.

$$\begin{aligned}\text{Var}(X) &= \mathbb{E}((X-\mu)^2)\\ &= \mathbb{E}(X^2 - 2X\mu + \mu^2)\\ &= \mathbb{E}(X^2) - 2\mu\cdot\mathbb{E}(X) + \mu^2\\ &= \mathbb{E}(X^2) - \mu^2\end{aligned}$$

□

Theroem:

$$\text{Var}(aX+b) = a^2\text{Var}(X)$$
$$SD(aX+b) = |a|SD(X)$$

Proof. $Y = aX + b$, then $\mu_Y = a\mu_X + b$

$$\begin{aligned}\text{Var}(Y) &= \mathbb{E}((Y-\mu_Y)^2)\\ &= \mathbb{E}([aX+b-(a\mu_X+b)]^2)\\ &= \mathbb{E}([aX-a\mu_X]^2)\\ &= \mathbb{E}(a^2(X-\mu_X)^2)\\ &= a^2\mathbb{E}((X-\mu)^2)\\ &= a^2\text{Var}(X)\end{aligned}$$

We known that $SD(x) = \sqrt{\text{Var}(X)}$, we square root both side, then:

$$SD(X) = \sqrt{a^2\text{Var}(X)}$$
$$SD(x) = |a|\sqrt{\text{Var}(X)}$$

□

1.2 Moment Generating Function

Moment Generating Function:

$$M(t) = \mathbb{E}(e^{tx}) = \sum_{x\in\mathbb{A}} e^{tx} f(x)$$

Example $f(X) = \frac{12}{35(X+2)}$, for x = -1, 0, 1, 2. Find mgf.

$$M(t) = \mathbb{E}(e^{tx}) = \sum_{x\in\mathbb{A}} e^{tx} f(x) = \sum_{x\in\mathbb{A}} \frac{12}{25(X+2)}$$

$$\begin{aligned} M(t) &= \frac{12}{25}e^{-t} + \frac{6}{25}e^{0} + \frac{4}{25}e^{t} + \frac{3}{25}e^{2t} \\ &= \frac{6}{25} + \frac{12}{25}e^{-t} + \frac{4}{25}e^{t} + \frac{3}{25}e^{2t} \end{aligned}$$

Example:

Theorem: Uniqueness Theorem of mgf Suppose random variable X, Y have $M_X(t), M_Y(t)$. If $M_X(t) = M_Y(t)$ for all $t \in (-a, a)$, for $a > 0$. Then X, Y have same probability distribution
The quantity of $\mathbb{E}(X^n), n = 1, 2, 3, ...$, is called n^{th} moment of probability distribution, or n^{th} moment to X.

Theorem: If x is a random variable with $mfgM(t)$, then

$$\mathbb{E}(X^n) = M^{(n)}(0)$$

Proof. Assume existence $M(t) = \sum_{\text{all } x} e^{tx} f(x)$ for $t \in (-a, a)$

$$\begin{aligned} M^{(n)}(t) &= \frac{\mathrm{d}^n}{\mathrm{d}t^n}(\sum_{x\in\mathbb{A}}^{\infty} e^{tx} f(x)) \\ &= \sum_{\text{all } x} \frac{\mathrm{d}^n}{\mathrm{d}t^n} e^{tx} f(x) \\ &= \sum_{\text{all } x} (x^n)(e^{tx}) f(x) \end{aligned}$$

Let $t = 0$, then all the $e^{tx} = 1$, which implies:

$$\begin{aligned} M^{(n)}(0) &= \left.\frac{\mathrm{d}^n M(t)}{\mathrm{d}t^n}\right|_{t=0} \\ &= \sum_{\text{all } x} (x^n)(e^{tx}) f(x) \\ &= \sum_{\text{all } x\in\mathbb{A}} x^n f(x) \\ &= \mathbb{E}(x^n) \end{aligned}$$

□

Example: $f(X) = (\frac{1}{2})^X$, calculate mean and variance of X. Use $\mathbb{E}(X^n) = \frac{\mathrm{d}^n M(0)}{\mathrm{d}t^n}$.

$$\begin{aligned} \mu &= \frac{\mathrm{d}}{\mathrm{d}t}(M(0)) \\ &= \left[\frac{e^t}{2}(1 - \frac{e^t}{2})^{-1} + \frac{e^{2t}}{4}(1 - \frac{e^t}{2})^{-2}\right]_{t=0} \\ &= \frac{1}{2}(1 - \frac{1}{2})^{-1} + \frac{1}{4}(1 - \frac{1}{2})^{-2} \\ &= 2 \end{aligned}$$

$$\begin{aligned}\mathbb{E}(X^2) &= M^{(2)}(0)\\ &= \left[\frac{e^{2t}}{4} - 2(1-\frac{e^t}{2})^{-3} + \frac{2e^{2t}}{4}(1-\frac{e^t}{2})^{-2} + \frac{e^t}{2}(1-\frac{e^t}{2})^{-1}\right]_{t=0}\\ &= \frac{1}{4}(1-\frac{1}{2})^{-3} + \frac{3}{4}(1-\frac{1}{2})^{-2} + \frac{1}{2}(1-\frac{1}{2})^{-1}\\ &= 6\end{aligned}$$

$$\mathrm{Var}(X) = \mathbb{E}((X-\mu)^2) = \mathbb{E}(X^2) - \mathbb{E}(X)^2 = 6 - 2^2 = 2$$

Theorem: Let X be random variable with $mgfM_X(t)$. If a and b are real constant such that $Y = aX + b$, them Y has mgf given by $M_Y = e^{bt}M_X(at)$.

Proof. Since $Y = aX + b$,

$$\begin{aligned}M_Y &= \mathbb{E}(e^{tY})\\ &= \mathbb{E}(e^{t(aX+b)})\\ &= \mathbb{E}(e^{atX+bt})\\ &= \mathbb{E}(e^{atX})\mathbb{E}(e^{bt})\\ &= e^{bt}\mathbb{E}(e^{atX})\\ &= e^{bt}M_X(at)\end{aligned}$$

□

1.3 Special Discrete Probability Distribution

Discrete Uniform Distribution: X takes on values in the set $A = \{a, a+1, ..., b\}$ such that $a \le b$. Suppose that all values are equal likely.

$$X \sim DU(a, b)$$

Probability Distribution: $b - a + 1$ values X with equal probability.

$$f(X) = \frac{1}{b-a+1} \quad \text{for } x = a, a+1, ..., b$$

1.4 Expectation, Covariance, Correlation

let $U = g(X, Y)$, then $f_U(u) = P(U = u) = \sum_{g(X,Y)=u} f(X, Y)$

$$\mathbb{E}(U) = \sum_{all(X,Y)} g(X, Y), f(X, Y)$$

Extended to n random variables, $X_1, X_2, \ldots, X_n, U = g(X_1, X_2, \ldots, X_n)$.

$$\mathbb{E}(U) = \sum_{all(X_1,X_2,...,X_n)} g(X_1, X_2, \ldots, X_n) f(X_1, X_2, \ldots, X_n)$$

Example: Suppose that $(X_1, X_2, \ldots, X_n) \sim MN(n, p_1, \ldots, p_k)$. For i,j $\in 1, 2, \ldots, k$ and $i < j$.Determine $\mathbb{E}(X, Y)$

$$f_{ij}(X_i, X_j) = \frac{n!}{x_i!x_j!(n - x_j - x_j)!}(p_i)^{x_i}(p_j)^{x_j}$$

$$\begin{aligned}\mathbb{E}(X_iX_j) &= \sum_{all(x_1,\ldots,x_n)} f(x_ix_j) \\ &= \sum_{all(x_1,\ldots,x_n)} x_ix_j(\frac{n!}{x_i!x_j!(n-x_i-x-j)!}(p_i)^{x_i}(p_j)^{x_j}(1-p_i-p_j)^{n-x_i-x_j})\end{aligned}$$

Let $Y_i = X_i - 1$ and $Y_j = X_j - 1$, then:

$$\begin{aligned}&= n(n-1)p_ip_j \sum_{all(y_i,y_j=0,1,\ldots,n,y_i+y_j\leq n-2)} \frac{(n-2)!}{y_i!y_j!((n-2)-y_i-y_j)}p_i^{y_i}p_j^{y_j}(1-p_i-p_j)^{((n-2)-y_i-y_j)} \\ &= n(n-1)p_ip_j(p_i+p_j+(1-p_i-p_j))^{n-2} \\ &= n(n-1)p_ip_j\end{aligned}$$

For $c_1, c_2, \ldots, c_n$ are real constants, then

$$\mathbb{E}(\sum_{i=1}^{n} c_ig_i(X_1, X_2, \ldots, X_n)) = \sum_{i=1}^{n} c_i\mathbb{E}(X_i)$$

Suppose that X and Y are independent random variables, then

$$\mathbb{E}(g_1(X)g_2(Y)) = \mathbb{E}(g_1(X))\mathbb{E}(g_2(Y))$$

Proof.

$$\begin{aligned}\mathbb{E}(g_1(X)g_2(Y)) &= \sum_{allX,Y} g_1(x)g_2(x)f(x,y) \\ &= \sum_{allX,Y} (g_1(X)f(x))(g_2(Y)f(y)) \\ &= \mathbb{E}(g_1(x))\mathbb{E}(g_2(y))\end{aligned}$$

□

Corollary:

$$\mathbb{E}(\prod_{i=1}^{n} g_i(X_i)) = \prod_{i=1}^{n} E(g_i(X_i))$$

Proof.

$$\begin{aligned}\mathbb{E}(\prod_{i=1}^{n} g_i(X_i)) &= \sum_{allX_i} \prod_{i=1}^{n} g_i(X_i) \\ &= \prod_{i=1}^{n} \sum_{allX_i} g_i(X_i) \\ &= \prod_{i=1}^{n} \mathbb{E}(g_i(X_i))\end{aligned}$$

□

Example: $X_1, \ldots, X_n$ be independent random variables where $X_i \sim Geo(p)$ for $i = 1, 2, \ldots, k$. Determin the probability distribution of $T = \sum_{i=1}^{k} X_i$

$$M_T(t) = \frac{p}{1-(1-p)e^t} t = 1, 2, \ldots, k, t < \ln(1-p)^{-1}$$

$$\begin{aligned} M_T(t) &= \frac{p}{1-(1-p)e^t}, \quad t = 1, 2, \ldots, k, \quad t < \ln((1-p)^{-1}) \\ &= \mathbb{E}(e^{tT}) \\ &= \mathbb{E}(e^{(t(X_1+X_2+\ldots X_k))}) \\ &= \mathbb{E}(e^{tX_1})\mathbb{E}(e^{tX_2}) \ldots \mathbb{E}(e^{tX_k}) \\ &= \prod_{i=1}^{k} M_T(t)(e^{tX_i}) \\ &= (\frac{p}{1-(1-p)e^t})^k \end{aligned}$$

Covariance: Random Variables X and Y,

$$\mathrm{Cov}(X, Y) = \mathbb{E}(XY) - \mu_X \mu_Y$$

Proof.

$$\begin{aligned} \mathrm{Cov}(X, Y) &= \mathbb{E}((X - \mu_X)(Y - \mu_Y)) \\ &= \mathbb{E}(XY - X\mu_Y - Y\mu_X + \mu_X\mu_Y) \\ &= \mathbb{E}(XY) - \mathbb{E}(X)\mu_Y - \mathbb{E}(Y)\mu_X + \mu_X\mu_Y \\ &= \mathbb{E}(XY) - \mu_X\mu_Y - \mu_X\mu_Y + \mu_X\mu_y \\ &= \mathbb{E}(XY) - \mu_X\mu_Y \end{aligned}$$

□

Special case when X and Y are independent random variables, then covariance equation becomes:

$$\mathrm{Cov}(X, Y) = \mathbb{E}(XY) - \mu_X\mu_Y = \mathbb{E}(X)\mathbb{E}(Y) - \mu_X\mu_Y = 0$$

Example: Suppose that $(X_1, X_2, \ldots, X_n) \sim MN(N; p_1, \ldots, p_n), i = 1, \ldots, k$ and $i < j$. Determine the $\mathrm{Cov}(X_i, Y_j)$.

$$\mathbb{E}(X_i X_j) = n(n-1)p_i p_j$$

$X_i \sim Bin(n, p_i) \Rightarrow \mathbb{E}(X_i) = np_i$ and $Y_j \sim Bin(n, p_j) \Rightarrow \mathbb{E}(Y_j) = np_j$.

$$\begin{aligned} \mathrm{Cov}(X_i X_j) &= \mathbb{E}(X_i X_j) - E(X_i)E(Y_j) \\ &= n(n-1)p_i p_j - n^2 p_i p_j \\ &= -np_i p_j \end{aligned}$$

Corollary:

$$\mathrm{Cov}(aX, bY) = ab\mathrm{Cov}(X, Y)$$

Proof.

$$\begin{aligned} \mathrm{Cov}(aX, bY) &= \mathbb{E}((aX)(bY)) - \mathbb{E}(aX)\mathbb{E}(bY) \\ &= ab\mathbb{E}(XY) - ab\mathbb{E}(X)\mathbb{E}(Y) \\ &= ab\mathrm{Cov}(X, Y) \end{aligned}$$

□

Correlation Coefficient ρ:

$$Corr = \frac{\text{Cov}(X,Y)}{\sigma_X \sigma_Y}$$

The range of ρ is $-1 \leq \rho \leq 1$

Proof.

We defined a new variable

$$Z = Y - tX$$

where t is a real number.

$$\begin{aligned}
\text{Var}(Z) &= \mathbb{E}((Z-\mu_Z)^2) \\
&= \mathbb{E}(((Y-tX)-(\mu_Y - t\mu_X))^2) \\
&= \mathbb{E}(((Y-\mu_Y)-t(\mu_X - X))^2) \\
&= \mathbb{E}((Y-\mu_Y)^2 - 2t(X-\mu_X)(Y-\mu_Y) + t^2(\mu_X - X)^2) \\
&= t^2\sigma_X^2 - 2\sigma_X\sigma_Y + \sigma_Y^2
\end{aligned}$$

Since $Var(Z) \geq 0$ for all $t \in \mathbb{R}$, then t at most have one real root.Use discriminant of this quadratic equation must be less or equal to 0.

$$\text{Cov}(X,Y)^2 - 4\sigma_X^2\sigma_Y^2 \leq 0$$

$$\left|\frac{\text{Cov}(X,Y)}{\sigma_X\sigma_Y}\right| \leq 1 \implies -1 \leq \rho \leq 1$$

□

1.5 Linear Combination of Random Variables

If $X_1, X_2, ..., X_n$ are n random variables such that $\mu_i = \mathbb{E}(X_i)$ and $\sigma_i^2 = \text{Var}(X_i)$ for $i = 1, 2, ..., n$, then we defined V be the linear combination of $c_i X_i$

$$V = \sum_{i=1}^{n} c_i X_i$$

where $c_1, c_2, ..., c_n$ are real constants.

$$\mathbb{E}(V) = \mathbb{E}(\sum_{i=1}^{n} c_i X_i) = \sum_{i=1}^{n} c_i \mu_i$$

Consider another sequence of random variables, $Y_1, Y_2, ...Y_r$ and define

$$W = \sum_{j=1}^{r} d_j Y_j$$

where $d_1, d_2, ..., d_r$ are real constants and $\alpha_j = \mathbb{E}(Y_j)$ for $j = 1, 2, ..., r$.

$$\begin{aligned}
\text{Cov}(V,W) &= \mathbb{E}((V-\mathbb{E}(V))(W-\mathbb{E}(W))) \\
&= \mathbb{E}\left(\left(\sum_{i=1}^{n} c_i(X_i - \mu_i)\right)\left(\sum_{j=1}^{r} d_j(Y_j - \alpha_j)\right)\right) \\
&= \sum_{i=1}^{n}\sum_{j=1}^{r} c_i d_j \mathbb{E}((X_i - \mu_i)(Y_j - \alpha_j)) \\
&= \sum_{i=1}^{n}\sum_{j=1}^{r} c_i d_j \text{Cov}(X_i Y_j)
\end{aligned}$$

Suppose that $n = r$, $X_i = X_j$ and $c_i = c_j$. We set $V = W$. Since $\text{Cov}(V, W) = \text{Var}(V)$.

$$\begin{aligned}
\text{Var}(V) &= \sum_{i=1}^{n}\sum_{j=1}^{n} c_i c_j \text{Cov}(X_i, X_j) \\
&= \sum_{i=j}^{n} c_i^2 \text{Var} X_i + \sum_{i=1}^{n}\sum_{j\neq i, j=1}^{n} \text{Cov}(X_i, X_j) \\
&= \sum_{i=j}^{n} c_i^2\sigma_i^2 + \sum_{i=1}^{n}\sum_{j=i+1}^{n} c_i c_j \text{Cov}(x_i, x_j) + \sum_{i=1}^{n}\sum_{j=1}^{i-1} c_i c_j \text{Cov}(x_i, X_j) \\
&= \sum_{i=j}^{n} c_i^2\sigma_i^2 + 2\sum_{i=1}^{n}\sum_{j=i+1}^{n} c_i c_j \text{Cov}(X_i, X_j)
\end{aligned}$$

For $n = 2$, then

$$\text{Var}(X_1 + X_2) = \sigma_1^2 + \sigma_2^2 + 2\text{Cov}(X_1, X_2)$$

$$\text{Cov}(X_1 + X_2 + \ldots X_n) = \mathbb{E}(X_1 X_2 \ldots X_n) - \mathbb{E}(X_1)\mathbb{E}(X_2), \ldots, \mathbb{E}(X_n) = 0$$

as $X_1, X_2, \ldots, X_n$ are independent random variables.Equation can be simplified as

$$\text{Var}(\sum_{i=1}^{n} X_i) = \sum_{i=1}^{n} \sigma_i^2$$

1.6 Conditional Probability Distribution

Suppose that X and Y are discrete random variables with joint $pmff(x, y)$.Let $f_X(x)$ be the marginal pmf of X and let $f_Y(y)$ be the marginal pmf of Y. The **conditional probaility mass function(pmf) of** X **given** $Y = y$ is

$$f_{X|Y}(x|y) = p(X = x|Y = y) = \frac{P(X = x|Y = y)}{P(Y = y)} = \frac{f(x, y)}{f_Y(y)}$$

provided that $f_Y(y) > 0$ Similarly, the **conditional** pmf **of** Y **given** $X = x$is

$$f_{Y|X}(y|x) = P(Y = y|X = x) = \frac{X = x, Y = y}{P(X = x)} = \frac{f(x, y)}{f_X(x)}$$

provided that $f_X(x) > 0$
for $f_{Y|X}(y|x) \geq 0$ and $f_{X|Y}(x|y) \geq 0$, moreover

$$\sum_{allx} f_{X|Y}(x|y) = 1 and \sum_{ally} f_{Y|X}(y|x) = 1$$

If X and Y are independent random variables, then $f(x, y) = f_X(x)f_Y(y)$ for all (x, y) implies that
$f_{X|Y}(x|y) = f_X(X) \text{and} f_{Y|X}(y|x) = f_Y(y)$ Expended to n random variables: Suppose that $X_1, X_2, \ldots X_n$ are discrete random variables with join $pmff(x_1, x_2, \ldots, x_n)$ We can define the joint pmf of $(X_i, \ldots, X_k)$, given $(X_j, \ldots, X_n)$ by

$$f_{X_i,\ldots,X_k|X_j,\ldots,X_n}(x_i, \ldots, x_k|x_j, \ldots, x_n) = P(X_i = x_i, \ldots, X_k = x_k|X_j = x_j, \ldots, X_n = x_n) = \frac{f(x_i, \ldots, f_n)}{f_{X_j,\ldots,X_n}(x_j, \ldots, x_n)}$$

Conditional mean of X given by Y:

$$\mathbb{E}(X|Y) = \sum_{allX} x f_{X|Y}(x|y)$$

If $g(X, Y)$ is a function of random variables X and Y, then we can define

$$\mathbb{E}(g(X, Y)|Y = y) = \sum_{allx} g(x, y) f_{X|Y}(x|y)$$

Suppose that $X_1, \ldots, X_n$ are discrete random variables with joint $pmf f(x_1, \ldots, x_n)$.

$$\mathbb{E}(\sum_{i=1}^{n}(X_i)) = \sum_{all x_1} \cdots \sum_{all x_n} (\sum_{i=1}^{n} X_i) f_{12\ldots(n-1)n}(x_1, x_2, \ldots, x_{n-1}, x_n)$$

If $c_1, \ldots, c_n$ are real constants, and $g_1, \ldots, g_n$ are arbitrary real valued functions, then

$$\mathbb{E}(\sum_{i=1}^{n} c_i g_(X_i)|Y = y) = \sum_{i=1}^{n} c_i \mathbb{E}(g_i(X_i)|Y = y)$$

Proof.

$$\begin{aligned}
\mathbb{E}\left(\sum_{i=1}^{n} c_i g_i(X_i) \,\middle|\, Y = y\right) &= \sum_{\text{all } x_1} \cdots \sum_{\text{all } x_n} \left[\left(\sum_{i=1}^{n} c_i g_i(X_i)\right) f_{X_1 \ldots X_n|Y}(X_1, \ldots, X_n|Y)\right] \\
&= \sum_{\text{all } x_1} \cdots \sum_{\text{all } x_n} \left[\sum_{i=1}^{n} c_i g_i(X_i) \frac{f(X_1, \ldots, X_n, Y)}{f_Y(y)}\right] \\
&= \sum_{i=1}^{n} \frac{c_i}{f_Y(y)} \left[\sum_{\text{all } x_i} g_i(X_i) f(x_1, \ldots, x_n, Y)\right] \\
&= \sum_{i=1}^{n} \frac{c_i}{f_Y(y)} \sum_{\text{all } x_i} g_i(X_i) f_{X_i,Y}(X_i, Y) \\
&= \sum_{i=1}^{n} c_i \left[\sum_{x_i} g_i(X_i) f_{X_i|Y}(x_i|y)\right] \\
&= \sum_{i=1}^{n} c_i \mathbb{E}\left(g_i(X_i) \,|\, Y = y\right)
\end{aligned}$$

□

Example: $X \sim Poi(\mu)$ and Y given $X = x$, follows $Bin(x, p)$ distribution. Find conditional pmf of X given $Y = y$.

$$f_{X|Y}(x, y) = \frac{P(X = x, Y = y)}{P(Y = y)}$$

$$P(Y = y|X = x) = \frac{P(X = x, Y = y)}{P(X = x)}$$

Which implies that

$$P(X = x|Y = y) = P(X = x)P(Y = y|X = x)$$

Since $X \sim Poi(\mu)$ and $Y \sim Bin(x, p)$, we get

$$P(X = x|Y = y) = \frac{e^{-\mu}\mu^x}{x!}\binom{x}{y} p^y (1 - p)^{x-y}$$

$$\begin{aligned}
P(Y = y) &= \sum_{\text{all } x} P(X = x, Y = y) \\
&= \sum_{x=y}^{\infty} \left(e^{-\mu}\frac{\mu^x}{x!}\binom{x}{y} p^y (1 - p)^{x-y}\right) \\
&= \frac{e^{-\mu}(\mu p)^y}{y!} \sum_{x=y}^{\infty} \frac{(\mu(1 - p))^{x-y}}{(x - y)!}
\end{aligned}$$

Let $z = x - y$, then the summation becomes:

$$\begin{aligned}
&= \frac{e^{-\mu}(\mu p)^y}{y!} \sum_{z=0}^{\infty} \frac{(\mu(1-p))^z}{z!} \\
&= \frac{e^{-\mu}(\mu p)^y}{y!} e^{\mu(1-p)} \\
&= \frac{e^{-\mu p}(\mu p)^y}{y!}
\end{aligned}$$

Substitute $P(X = x|Y = y)$ and $P(Y = y)$ into original equation.

$$\begin{aligned}
f_{X|Y}(x|y) &= \frac{P(X = x, Y = y)}{P(Y = y)} \\
&= \frac{\frac{e^{-\mu}\mu^x}{x!}\binom{x}{y}p^y(1-p)^{x-y}}{\frac{e^{-\mu p}(\mu p)^y}{y!}} \\
&= \frac{e^{-\mu(1-p)}(\mu(1-p))^{x-y}}{(x-y)!}
\end{aligned}$$

1.7 Law of Total Expectation

$$E(g(X)|Y) = E(v(Y)) = \sum_{ally} v(y) f_Y(y) = \sum_{ally} E(g(X)|Y = y) f_Y(y)$$

Theorem: **Law of Total Expectation**: Let g be arbitrary real valued function. For random variables X and$Y$.

$$E(g(X)) = E(E(g(X)|Y))$$

Proof.

$$\begin{aligned}
E(E(g(X)|Y)) &= \sum_{ally} E(g(X)|Y = y) f_Y(y) \\
&= \sum_{ally} \left(\sum_{allx} g(X) \frac{f(x,y)}{f_Y(y)} \right) f_Y(y) \\
&= \sum_{allx} g(X) \sum_{ally} f(x,y) \\
&= \sum_{allx} g(X) f_X(x) \\
&= E(g(X))
\end{aligned}$$

□

Example: Let Y have the probability distribution:

y	1	2	3
$f_Y(y)$	1/2	1/4	1/4

Suppose that $X|Y = y \sim \text{Poi}(\mu_y)$, where $\mu_1 = 9.4$, $\mu_2 = 11.9$, and $\mu_3 = 13.5$. Find $\mathbb{E}(X)$ and $\text{Var}(X)$.

Proof. By the Law of Iterated Expectations:

$$\begin{aligned}\mathbb{E}(X) &= \sum_{y=1}^{3} \mathbb{E}(X|Y=y) f_Y(y) \\ &= 9.4\left(\frac{1}{2}\right) + 11.9\left(\frac{1}{4}\right) + 13.5\left(\frac{1}{4}\right) \\ &= 11.05\end{aligned}$$

To find $\text{Var}(X)$, we first calculate $\mathbb{E}(X^2)$ using the Law of Iterated Expectations:

$$\begin{aligned}\mathbb{E}(X^2) &= \sum_{y=1}^{3} \mathbb{E}(X^2|Y=y) f_Y(y) \\ &= \mathbb{E}(W_1^2)\left(\frac{1}{2}\right) + \mathbb{E}(W_2^2)\left(\frac{1}{4}\right) + \mathbb{E}(W_3^2)\left(\frac{1}{4}\right)\end{aligned}$$

Since $\text{Var}(W) = \mathbb{E}(W^2) - (\mathbb{E}(W))^2$, we have $\mathbb{E}(W^2) = \text{Var}(W) + (\mathbb{E}(W))^2$:

$$\begin{aligned}\mathbb{E}(X^2) &= \left(\text{Var}(W_1) + \mu_1^2\right)\left(\frac{1}{2}\right) + \left(\text{Var}(W_2) + \mu_2^2\right)\left(\frac{1}{4}\right) + \left(\text{Var}(W_3) + \mu_3^2\right)\left(\frac{1}{4}\right) \\ &= (9.4 + 9.4^2)\left(\frac{1}{2}\right) + (11.9 + 11.9^2)\left(\frac{1}{4}\right) + (13.5 + 13.5^2)\left(\frac{1}{4}\right) \\ &= 97.76\left(\frac{1}{2}\right) + 153.51\left(\frac{1}{4}\right) + 195.75\left(\frac{1}{4}\right) \\ &= 136.195\end{aligned}$$

Finally, calculating the variance:

$$\begin{aligned}\text{Var}(X) &= \mathbb{E}(X^2) - (\mathbb{E}(X))^2 \\ &= 136.195 - (11.05)^2 \\ &= 14.0925\end{aligned}$$

□

1.8 Expectation, Covariance and Correlation

$$\mathbb{P}(U = u) = \sum_{\substack{\text{all } (x,y): \\ g(x,y)=u}} f(x,y)$$

$$\mathbb{E}(U) = \mathbb{E}(g(X,Y)) = \sum_{\text{all } (x,y)} g(x,y) f(x,y)$$

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n} (x_i - \bar{x})^2$$

2 Univariate Continuous Probability Distribution

Continuous Random Variable: $P(X = x) = 0$ The properties of a *cdf* are the same for continuous random variables as for discrete random variables. These properties include:

- $F(x)$ is defined for all $x \in \mathbb{R}$,
- $F(x)$ is a non-decreasing function of x for all $x \in \mathbb{R}$,
- $\lim_{x->-\infty} F(x) = 0$ and $lim_{x->\infty} F(x) = 1$
- $P(a < x \leq b) = F(b) - F(a)$

Proof. Using property (4), for any $a \in \mathbb{R}$

$$P(X = a) = \lim_{\epsilon->0} P(a - \epsilon < X \leq a) = \lim_{\epsilon->0} (F(a) - F(a - \epsilon)) = F(a) - \lim_{\epsilon->0} F(a - \epsilon) = 0$$

□

Suppose that X is continuous random variable withh *cdf* $F(x)$. For $p \in (0, 1)$, the $100p^{\text{th}}$ **percentile** of X is the value $q(o)$ such that $F(q(p)) = p$. If $p = 0, 5$, then $m = q(0.5)$ is called the **median** of X

2.1 Functions of Random Variables

Example: Let$X$ be a continuous random variable with *pdf* $f_X(x)$ and *cdf* $F_X(x)$. Determine a general formula for the *pdf* of $Y = X^2$ Since $Y = X^2$, then Y is non-negative continuous function and $y > 0$.

$$F_Y(y) = P(Y \leq y) = P(X^2 \leq y) = P(-\sqrt{y} \leq X \leq \sqrt{y}) = F_X(\sqrt{y}) - F_X(-\sqrt{y})$$

$$\begin{aligned} f_Y(y) &= F'_Y(y) \\ &= F'_X(\sqrt{y})\frac{d}{dy}(y^{\frac{1}{2}}) - F'_X(-\sqrt{y})\frac{d}{dy}(-y^{\frac{1}{2}}) \\ &= f_X(\sqrt{y}\frac{1}{2\sqrt{y}}) - f_X(-\sqrt{y})(\frac{-1}{2\sqrt{y}}), \text{for } y > 0 \end{aligned}$$

Theorem: Let X be continuous random variable with *pdf* $f_X(x)$.Suppose that g is a strictly monotonic differentiable function an range of X. Then, *pdf*$Y = g(X)$ in range where $f_Y(y) > 0$ given by

$$f_Y(y) = f_X(g^{-1}(y))|\frac{dg^{-1}(y)}{dy}|$$

Proof. Assume that g is monotonically increasing, then

$$F_Y(y) = P(Y \leq y) = P(g(X)\ ley) = P(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$$

$$f_Y(y) = F_Y(y) = \frac{d}{dy}(F_X(g^{-1}(y)) = F'_X(g^{-1}(y)))\frac{dg^{-1}(y)}{dy} = f_X(g^{-1}(y))\frac{dg^{-1}(y)}{dy}$$

Since g is monotonically increasing, g^{-1} is also monotonically increasing.

$$|\frac{dg^{-1}(y)}{dy}| = \frac{dg^{-1}(y)}{dy}$$

Assume that g is monotonically decreasing, Then

$$F_Y(y) = P(Y \leq y) = P(g(X) \leq y) = P(X \geq g^{-1}(y)) = 1 - F_X(g^{-1}(y))$$

$$f_Y(y) = F'_Y(y) = \frac{d}{dy}[1 - F_X(g^{-1}(y))] = F'_X(g^{-1}(y))\frac{dg^{-1}(y)}{dy} = f_X(g^{-1}(y))\frac{dg^{1}(y)}{dy}$$

Since g is monotonically decreasing, g^{-1} is also monotonically decreasing that implies that the derivatives are negative.

$$-\frac{dg^{-1}(y)}{dy} = |\frac{dg^{-1}(y)}{dy}|$$

□

Corollary: Let X be a continuous random variable with *pdf* $f_X(x)$. If a and b are real constants such that $a \neq 0$, then $Y = aX + b$ has *pdf* given by

$$f_Y(y) = \frac{1}{|a|}f_X(\frac{y-b}{a})$$

Proof. Define that $y = g(x) = ax + b$, then $x = \frac{y-b}{a}$, then by previous theorem,

$$f_Y(y) = f_X(g^{-1}(y))|\frac{dg^{-1}(y)}{dy}| = f_X(\frac{y-b}{a}|\frac{d}{dy}(\frac{y-b}{a})|) = \frac{1}{|a|}f_X(\frac{y-b}{a})$$

□

2.2 Expectation of Random Variable

The expected value or mean of continuous random variable X with *pdf* $f_X(x)$ is defined by

$$\mu_X = \mathbb{E}(X) = \int_{-\infty}^{\infty} x f_X(x)\,\mathrm{d}x$$

- *mgf* of X is defined as:

$$M_X(t) = \mathbb{E}(g(X)) = \int_{-\infty}^{\infty} g(x)f(x)\,\mathrm{d}x$$

  provided there exists all values of t within some open interval.

- The nth moment of X, where $n = 1, 2, \ldots$, is defined as:

$$E(X^n) = \int_{-\infty}^{\infty} x^n f_X(x)\,\mathrm{d}x$$

$$\mathbb{E}(X^n) = M_X(^)(n)(0) = \left.\frac{\mathrm{d}^n M_X(t)}{\mathrm{d}t^n}\right|_{t=0}, \text{for } n = 1, 2, 3, \ldots$$

- Variance of X id defined as:

$$\sigma_X^2 = \mathrm{Var}(X) = \mathbb{E}((x - \mu_x)^2) = \int_{-\infty}^{\infty}(x - \mu_k)^2\,\mathrm{d}x = \mathbb{E}(X^2) - \mathbb{E}(X)^2$$

$$\sigma_X = \mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}$$

- For real constant a and b.

$$\mathbb{E}(aX + b) = a\mu_X + b$$
$$\mathrm{Var}(aX + b) = a^2\sigma_X^2$$
$$\mathrm{SD}(aX + b) = |a|\,\sigma_X$$
$$M_{aX+b}(t) = e^{bt}M_{aX}(t)$$

2.3 Special Continuous Probability Distributions

2.3.1 Continuous Uniform Distribution

Consider a random variable X which takes on values in interval (a, b) where a and b are real numbers such that $a < b$. If the probability distribution of X satisfies all subintervals of (a, b) with the same length are equally likely, then X is said to have a **continuous uniform distribution**. Denote as $X \sim U(a, b)$

$$\int_a^b f(x)\,\mathrm{d}x = \int_a^b k\,\mathrm{d}x = k(b-a) \Rightarrow k = \frac{1}{b-a}$$

$$F(X) = \int_a^{\infty} \frac{1}{b-a}\,\mathrm{d}x = \frac{x-a}{b-a}$$

from equation $F(q(p)) = p$

$$\frac{q(p)-a}{b-a} = p \Rightarrow q(p) - a = p(b-a) \Rightarrow = q(p) = a(p-1) + bp$$

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} x(fx)\,\mathrm{d}x = \frac{1}{b-a}\frac{x^2}{2}\Big|_b^a = \frac{b^2-a^2}{2(b-a)} = \frac{(b-a)(b+a)}{2(b-a)} = \frac{b+a}{2}$$

$$\begin{aligned}
\mathrm{Var}(X) &= \mathbb{E}(X^2) - E(X)^2 \\
&= \int_a^b x^2 \frac{1}{b-a}\,\mathrm{d}x - \frac{b+a}{2}^2 \\
&= \frac{1}{b-a}\,\frac{x^3}{3}\Big|_a^b - \frac{b^2-2ab-a^2}{4} \\
&= \frac{1}{b-a}\frac{(b-a)(b^2+ab+a^2)}{3} - \frac{b^2+2ab+a^2}{4} \\
&= \frac{b^2-2ab+a^2}{12} \\
&= \frac{b-a^2}{12}
\end{aligned}$$

Theorem:Suppose that X is continuous random variable with *cdf* $F_X(X)$. Suppose that $F_X(X)$ is strictly increasing in following case: If $x_1 < x_2$ such that $0 < F_X(x_1)$ and $F_X(x_2) < 1$, then $F_X(x_1) < F_X(x_2)$ If $Y = F_X(X)$, then $Y \sim U(0, 1)$

Proof. If □

2.3.2 Exponential Distribution

Denote as $X \sim Exp(\theta)$

$$f(x) = \begin{cases} \frac{1}{\theta}e^{\frac{-X}{\theta}} & \text{for} x > 0, \\ 0 & \text{otherwise} \end{cases}$$

$$\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = \int_{-\infty}^{\infty} \frac{1}{\theta}e^{\frac{-x}{\theta}}\,\mathrm{d}x = -e^{\frac{-x}{\theta}}\Big|_0^{\infty} = \lim_{x->\infty} -e^{\frac{-x}{\theta}} + 1 = 1$$

$$F(X) = \int_0^x \frac{1}{\theta}e^{\frac{-z}{\theta}}\,\mathrm{d}z = -e^{\frac{-z}{\theta}}\Big|_0^x = 1 - e^{\frac{-x}{\theta}}$$

$$F(X) = \begin{cases} 0 & \text{otherwise} \\ 1 - e^{\frac{-x}{\theta}} & \text{for} x > 0 \end{cases}$$

Determine the p-quantile, from the equation $F(q(p)) = p$:

$$1 - e^{\frac{-q(p)}{\theta}} = p \Rightarrow \frac{-q(p)}{\theta} = -\ln(p-1) \Rightarrow q(p) = -\theta\ln(1-p)$$

$$\begin{aligned}
M(t) = \mathbb{E}(e^{tx}) &= \int_0^\infty e^{tx}\frac{1}{\theta}e^{\frac{-x}{\theta}}\,\mathrm{d}x \\
&= \frac{1}{\theta}\int_0^\infty e^{x(t-\frac{1}{\theta})}\,\mathrm{d}x \\
&= \frac{1}{\theta(t-\frac{1}{\theta})}\ \frac{1}{t-\frac{1}{\theta}}e^{x(t-\frac{1}{\theta})}\Big|_0^\infty \\
&= \frac{1}{t\theta-1}[\lim_{x->\infty} e^{x(t-\frac{1}{\theta})}-1] \\
&= \frac{1}{t\theta-1}[0-1] \qquad \text{provided } t-\frac{1}{\theta}<0 \Rightarrow t<\frac{1}{\theta} \\
&= \frac{1}{1-t\theta}
\end{aligned}$$

$M_1(t) = \theta(1-t\theta)^{-2}$
$M_2(t) = -2\theta(1-t\theta)^{-3}-\theta = 2\theta^2(1-t\theta)^{-3}$

$$\mathbb{E}(X) = M_1(0) = \theta$$
$$\mathrm{Var}(X) = \mathbb{E}(X^2) - E(X)^2 = M_2(0) - \theta^2 = 2\theta^2 - \theta^2 = \theta^2$$

2.3.3 Gaussian Distribution

$$f(X) = \frac{1}{\sqrt{2\pi}\sigma}e^{\frac{-1}{2}(\frac{x-\mu}{\sigma})^2}\text{for} -\infty < x < \infty$$

where $\mu \in \mathbb{R}$ and $\sigma > 0$ are parameter of probability distribution. To verify the area under the curve of $f(X)$ equal 1.

$$\begin{aligned}
\int_{-\infty}^\infty \frac{1}{\sqrt{2\pi}\sigma}e^{\frac{-1}{2}(\frac{x-\mu}{\sigma})^2}\,\mathrm{d}x &= \int_{-\infty}^\infty \frac{1}{\sqrt{2\pi}}e^{\frac{-y^2}{2}}\,\mathrm{d}y \qquad \text{if } y = \frac{x-\mu}{\sigma} \Rightarrow dy = \frac{dx}{\sigma} \\
&= \frac{1}{\sqrt{2\pi}}\cdot 2\int_0^\infty e^{\frac{-y^2}{2}}\,\mathrm{d}y \qquad \text{Since } e^{\frac{-y^2}{2}} \text{is an even function} \\
&= \frac{\sqrt{2}}{\pi}\int_0^\infty e^{\frac{-y^2}{2}}\,\mathrm{d}y
\end{aligned}$$

$$\begin{aligned}
\left(\frac{\sqrt{2}}{\sqrt{\pi}}\int_0^\infty e^{\frac{-y^2}{2}}\,\mathrm{d}y\right)^2 &= \frac{2}{\pi}\left(\int_0^\infty e^{\frac{-y^2}{2}}\,\mathrm{d}y\int_0^\infty e^{\frac{-z^2}{2}}\,\mathrm{d}z\right) \\
&= \frac{2}{\pi}\left(\int_0^\infty\int_0^\infty e^{-(y^2+z^2)}\,\mathrm{d}z\,\mathrm{d}y\right) \\
&= \frac{2}{\pi}\int_0^\infty re^{\frac{-r^2}{2}}\,\mathrm{d}r \qquad \text{Let } y = r\cos\theta \text{and } z = r\sin\theta \Rightarrow dydz = rdrd\theta \\
&= \int_0^\infty e^{-s}\,\mathrm{d}s \qquad \text{Let} s = \frac{r^2}{2} \qquad\qquad = -e^{-s}\big|_0^\infty = 1
\end{aligned}$$

Standard Nornal Distribution:

$$f_Z(Z) = \frac{1}{\sqrt{2\pi}}e^{\frac{-z^2}{2}} \qquad \text{for} -\infty < z < \infty, Z \sim N(0,1)$$

mgf:

$$\begin{aligned} M_Z(t) &= \int_{-\infty}^{\infty} e^{tz} \frac{1}{\sqrt{2\pi}} e^{\frac{-z^2}{2}} \, \mathrm{d}z \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{\frac{-1}{2}(z^2-2tz)} \, \mathrm{d}z \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{\frac{-1}{2}(z-t)^2} e^{t^2} \, \mathrm{d}z \quad \text{Since } \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{\frac{-1}{2}(z-t)^2} \, \mathrm{d}z \text{is } Z \sim N(t,1) \\ &= e^{t^2} \quad \text{for } t \in \mathbb{R} \end{aligned}$$

$$\mathbb{E}(Z) = \frac{d}{dz}(e^{\frac{t^2}{2}})\Big|_{z=0} = te^{\frac{t^2}{2}} = 0$$

$$\begin{aligned} \mathrm{Var}(Z) &= \mathbb{E}(Z^2) - \mathbb{E}(Z)^2 \\ &= \frac{\mathrm{d}^2}{\mathrm{d}z^2}\left[e^{\frac{t^2}{2}} + t^2 e^{\frac{t^2}{2}}\right]\Big|_{t=0} \\ &= 1 \end{aligned}$$

Theorem: If $X \sim N(\mu, \sigma^2)$ and $z = \frac{x-\mu}{\sigma}$, then $z \sim N(0,1)$.

Proof. Express $z = ax + b$ for $a = \frac{1}{\sigma}$ and $b = \frac{-\mu}{\sigma}$ By

$$f_Y(y) = \frac{1}{|a|} f_X(\frac{y-b}{a})$$

$$\begin{aligned} f_Z(z) &= \frac{1}{\left|\frac{1}{\sigma}\right|} f_X(\frac{z+\frac{\mu}{\sigma}}{\frac{1}{\sigma}}) \\ &= \sigma f_X(z\sigma + \mu) \quad \text{Since } X \sim N(\mu, \sigma^2) \\ &= \sigma \frac{1}{\sqrt{2\pi\sigma}} e^{\frac{-1}{2}(\frac{\sigma z+\mu-\mu}{\sigma})^2} \\ &= \frac{1}{\sqrt{2\pi}} e^{\frac{-1}{2}z^2} \quad \text{for} -\infty < z < \infty, \text{ such that} z \sim N(0,1) \end{aligned}$$

□

From previous theorem, $z = \frac{x-\mu}{\sigma}$, $z \sim N(0,1)$, if $X \sim N(\mu, \sigma^2)$

$$M_X(t) = M_{\sigma Z+\mu}(t) = e^{\mu t} M_Z(\sigma t) = e^{ut} \cdot e^{\frac{\sqrt{(\sigma t)}}{2}} \quad \text{for} t \in \mathbb{R}$$

$$\mathbb{E}(X) = E(\sigma Z + \mu) = \sigma \mathbb{E}(Z) + \mu = \sigma \cdot 0 + \mu = \mu$$

$$\mathrm{Var}(X) = \mathrm{Var}(\sigma Z + \mu) = \sigma^2 \mathrm{Var}(Z) + 0 = \sigma^2 \cdot 1 = \sigma^2$$

2.3.4 Central Limit Theorem(CLT)

Theorem: (Central Limit Theorem(CLT)):Let$X_1, \ldots, X_n$ be independent random variables that all have same probability distribution, with μ and σ^2.Let $S_n = \sum_{i=1}^{n} X_i$. Then as $n- > \infty$, the *cdf* of the random variable

$$\frac{S_n - n\mu}{\sigma\sqrt{n}}$$

approaches the $N(0,1)$*cdf*.

Sample Mean:

$$\bar{X_n} = \frac{1}{n}\sum_{i=1}^{n} X_i = \frac{S_n}{n}, \text{substitute into equation} \frac{S_n - n\mu}{\sigma\sqrt{n}}$$

$$\frac{S_n - n\mu}{\sigma\sqrt{n}} = \frac{n\bar{X_n} - n\mu}{\sigma\sqrt{n}} = \frac{\bar{X_n} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Remark: Apply CLT when n is large and finite to approximate the proability distribution of $\sum_{i=1}^{n} X_i$ or $\bar{X}$ by normal distribution.

$$S_n = \sum_{i=1}^{n} X_i \sim N(n\mu, n\sigma^2) \text{ and } \bar{X_n} \sim N(\mu, \frac{\sigma^2}{n})$$

3 Introduction to Statistical Science

Variate: A varaite is a characteristic of a unit. Attribute: An attribute of a population or process is a function of the variate over the population or process. Three types of empirical studies:

- Sample Survey: Obtained information by selecting a "representative" sample of units form the population and determining the variates of interest for each unit in the sample.
- Observational Studies: An observational study is one in which data are collected about a population or process without any attempt to change the value of one or more variates for the sampled units.
- Experimental Studies: An experimental study is one in which the experimenter (that is, the person conducting the study) intervenes and changes or sets the values of one or more variates for the units in the sample.

Data Summaries:

Sample Mean: $y = \frac{1}{n}\bar{\sum}_{i=1}^{n} X_i$

Sample Median: $\hat{m}$: the meddle value when n is odd and the sample is ordered from smallest to largest and the average of the two middle values when n is even.

Sample Mode: the value of y which have highest frequency appears in the sample.

Measure of dispersion or variability

Sample Variance:

$$s^2 = \frac{1}{n-1}\left[\sum_{i=1}^{n} {y_i}^2 - \frac{1}{n}\left(\sum_{i=1}^{n} y_i\right)^2\right]$$

And the sample standard deviation: $s = \sqrt{s^2}$

Proof.

$$\begin{aligned}
s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2 &= \frac{1}{n-1}\left[\sum_{i=1}^{n} {y_i}^2 - 2y_i\bar{y} - \bar{y}^2\right] \\
&= \frac{1}{n-1}\left[\left(\sum_{i=1}^{n} {y_i}^2\right) - 2\bar{y}\left(\sum_{i=1}^{n} y_i\right) + \sum_{i=1}^{n}\bar{y}^2\right] \qquad \text{Since } \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i \\
&= \frac{1}{n-1}\left[\left(\sum_{i=1}^{n} {y_i}^2\right) - 2n\bar{y}^2 - n\bar{y}^2\right] \\
&= \frac{1}{n-1}\left[\left(\sum_{i=1}^{n} {y_i}^2\right) - n\left(\sum_{i=1}^{n} y_i\right)^2\right] \\
&= \frac{1}{n-1}\left[\sum_{i=1}^{n} {y_i}^2 - \frac{1}{n}\sum_{i=1}^{n} y_i^2\right]
\end{aligned}$$

□

Expectation of Sample Mean:

$$\mathbb{E}(\bar{X}) = \mathbb{E}(X)$$

Proof.

$$\mathbb{E}(\bar{X}) = \mathbb{E}(\frac{1}{n}\sum_{i=1}^{n} X_i) = \frac{1}{n}\sum_{i=1}^{n}(\mathbb{E}(X_i)) = \frac{1}{n}n\mathbb{E}(X) = \mathbb{E}(X)$$

□

Variance of Sample Mean:

$$\mathrm{Var}(\bar{X}) = \frac{\mathrm{Var}(X)}{n}$$

Proof.

$$\begin{aligned}
\mathrm{Var}(\bar{X}) &= \mathrm{Var}(\frac{1}{n}\sum_{i=1}^{n} X_i) \\
&= \frac{1}{n^2}\mathrm{Var}(\sum_{i=1}^{n} X_i) \\
&= \frac{\sum_{i=1}^{n}\mathrm{Var}(X_i)}{n^2} \\
&= \frac{n\mathrm{Var}(X)}{n^2} \\
&= \frac{\mathrm{Var}(X)}{n}
\end{aligned}$$

□

The stnadard error is $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$
Expectation of Sample Variance:

$$\mathbb{E}({s_n}^2) = \mathrm{Var}(X)$$

Proof.

$$\begin{aligned}
\mathbb{E}(S_n^2) &= \mathbb{E}\left(\frac{1}{n-1}\sum_{i=1}^{n}\left(X_i-\bar{X}\right)^2\right)\\
&= \frac{1}{n-1}\mathbb{E}\left(\sum_{i=1}^{n}\left(X_i-\bar{X}\right)^2\right)\\
&= \frac{1}{n-1}\mathbb{E}\left(\sum_{i=1}^{n}(X_i^2-2X_i\bar{X}+\bar{X}^2)\right)\\
&= \frac{1}{n-1}\mathbb{E}\left(\sum_{i=1}^{n}X_i^2-2\bar{X}\sum_{i=1}^{n}X_i+\sum_{i=1}^{n}\bar{X}^2\right)\\
&= \frac{1}{n-1}\mathbb{E}\left(\sum_{i=1}^{n}X_i^2-2\bar{X}(n\bar{X})+n\bar{X}^2\right)\\
&= \frac{1}{n-1}\mathbb{E}\left(\sum_{i=1}^{n}X_i^2-n\bar{X}^2\right)\\
&= \frac{1}{n-1}\left(\sum_{i=1}^{n}\mathbb{E}(X_i^2)-n\mathbb{E}(\bar{X}^2)\right)\\
&= \frac{1}{n-1}\left(n(\mathrm{Var}(X)+(\mathbb{E}(X))^2)-n\left(\frac{\mathrm{Var}(X)}{n}+(\mathbb{E}(X))^2\right)\right)\\
&= \frac{1}{n-1}\left(n\mathrm{Var}(X)+n(\mathbb{E}(X))^2-\mathrm{Var}(X)-n(\mathbb{E}(X))^2\right)\\
&= \frac{1}{n-1}\left((n-1)\mathrm{Var}(X)\right)\\
&= \mathrm{Var}(X)
\end{aligned}$$

□

Measure of Shape:

- The sample skewness:
$$g_1 = \frac{\frac{1}{n}\sum_{i=1}^{n}\left(y_i-\bar{y}\right)^3}{\left[\frac{1}{n}\sum_{i=1}^{n}\left(y-\bar{y}\right)^2\right]^2}$$
- The sample kurtosis:
$$g_2 = \frac{\frac{1}{n}\sum_{i=1}^{n}\left(y_i-\bar{y}\right)^4}{\left[\frac{1}{n}\sum_{i=1}^{n}\left(y_i-\bar{y}\right)^2\right]^2}$$

Interquartile range: $IQR = q(0.75) - q(0.25)$
Five number summary: Smallest observation, the lower quartile, the median, the upper quartile, and the largest value. $y_{(1)}, q(0.25), q(0.5), q(0.75), y_{(n)}$.

Sample Correlation:The sample correlationfor data $(x_1, y_1), \ldots, (x_n, y_n)$

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} = \frac{\left(\sum_{i=1}^{n}\left(x_i-\bar{x}\right)\left(y_i-\bar{y}\right)\right)}{\left(\sum_{i=1}^{n}\left(x_i-\bar{x}\right)^2\right)\left(\sum_{i=1}^{n}\left(y_i-\bar{y}\right)^2\right)}$$

3.1 Point Estimate and Maximum Likelihood Estimation

A **point estimate** of a parameter is the value of a function of the observed data $y_1, \ldots, y_n$ and other known quantities, use $\bar{\theta}$ to denote an estimate of the parameter θ. Likelihood Function for Discrete Distribution:

$$L(\theta) = L(\theta; y) = P(Y = y; \theta) \qquad \text{for} \theta \in \Omega$$

where the parameter space Ω is the set of possible value for θ.

Maximize Likelihood Esitmation (MLE): If the value of θ which maximize the probability of observing the data y. Denote as $\bar{\theta}$. Example: $Y \sim Bin(n, \theta)$

$$L(\theta) = P(Y = y; \theta) = \binom{n}{y} \theta^y (1-\theta)^{n-y} \qquad \text{for} 0 \leq \theta \leq 1$$

Assume $y \neq 0$, $y \neq n$, let $\frac{dL(\theta)}{d\theta} = 0$.

$$\begin{aligned} 0 &= \binom{n}{y} \left[y\theta^{y-1}(1-\theta)^{n-y} + \theta^y (1-\theta)^{n-y-1}(-1)(n-y) \right] \\ &= \binom{n}{y} \theta^{y-1}(1-\theta)^{n-y-1}(y(1-\theta) - \theta(n-y)) \\ &\Rightarrow y - y\theta - n\theta + y\theta = 0 \\ &\Rightarrow y = n\theta \qquad \text{Sample Proportion} \end{aligned}$$

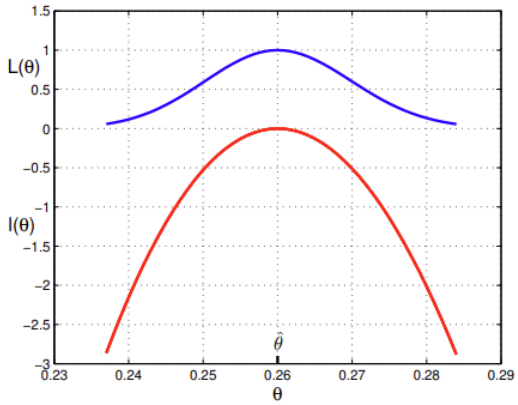
Relative Likelihood Function:

$$R(\theta) = \frac{L(\theta)}{L(\bar{\theta})} \qquad \text{for} \theta \in \Omega, 0 \leq R(\theta) \leq 1, \forall \theta \in \Omega$$

Log Likelihood Function:

$$l(\theta) = \ln L(\theta) = \log L(\theta) \qquad \text{for } \theta \in \Omega$$

Since the log function is a monotonw funciton of $L(\theta)$, two function increase and decrease toether. Both fucntions have a maximum at the same value $\theta = \hat{\theta}$



Example: Likelihood Function for Poisson Distirbution

$$\begin{aligned} L(\theta) &= \prod_{i=1}^{n} \frac{\theta^{y_i} e^{-\theta}}{y_i!} \\ &= \left(\prod_{i=1}^{n} \frac{1}{y_i!} \right) \theta^{\sum y_i} e^{-n\theta} \\ &\propto \theta^{n\bar{y}} e^{-n\theta} \quad \text{for } \theta \geq 0 \end{aligned}$$

$$\begin{aligned} l(\theta) &= \log(L(\theta)) = n\bar{y} \log \theta - n\theta + \text{constant} \\ \frac{dl(\theta)}{d\theta} &= \frac{n\bar{y}}{\theta} - n \\ \text{Set derivative to 0: } & \frac{n\bar{y}}{\hat{\theta}} - n = 0 \\ & \implies \hat{\theta} = \bar{y} \quad \text{maximizes } l(\theta) \end{aligned}$$

Combining Likelihood Based on Independent Experiments. Supposey_1 and y_2 statistically independent.

$$P(Y_1 = y_1, Y_2 = y_2; \theta) = P(Y_1 = y_1; \theta) P(Y_2 = y_2; \theta)$$

Likelihood Function $L(\theta)$ based on y_1 and y_2.

$$L(\theta) = Ł_1(\theta) L_2(\theta) \qquad \text{for } \theta \in \Omega, \text{where} L_i(\theta) = P(Y_i = y_i; \theta) \text{ for } i = 1, 2, \ldots$$

$Y \sim Bin(n, \theta)$ and Suppose $y_1, y_2, \ldots, y_n$ are independent random variables from Binomial distribution.

$$P(Y = y; \theta) = f(y; \theta) = \binom{n}{y} \theta^y (1 - \theta)^{\theta - y}$$

Ignoring all the constant term without θ.

$$L(\theta) = \theta^y (1 - \theta)^(n - y), \qquad \text{for } 0 \leq \theta \leq 1, f(y_i; \theta) = \theta^{y_i} (1 - \theta)^{1 - y_i}$$

$$\begin{aligned} L(\theta) &= \prod_{i=1}^{n} f(y_i, \theta) \\ &= \prod_{i=1}^{n} e^{y_i} (1 - \theta)^{1 - y_i} \\ &= \theta^{\sum_{i=1}^{n} y_i} (1 - \theta)^{\sum_{i=1}^{n} (1 - y_i)} \\ &= \theta^y (1 - \theta)^(n - y) \qquad \text{for } 0 \leq \theta \leq 1 \end{aligned}$$

$$l(\theta) = ln(L(\theta)) = ln\left[\theta^y (1 - \theta)^{n - y}\right] = y ln(\theta) + (n - y) ln(1 - \theta)$$

$$\frac{dl(\theta)}{d\theta} = \frac{y}{\theta}(n - y)\frac{-1}{1 - \theta} \Rightarrow y(1 - \theta) - \theta(n - y) = 0 \Rightarrow y - n\theta = 0$$

Therefore $\hat{\theta} = \frac{y}{n}$

3.2 Likelihood Function for Continuous Distribution

If $y_1, \ldots, y_n$ are observed value of random sample from distribution with probability density function $f(y_i, 0)$, then

$$L(\theta) = L(\theta, y) = \prod_{i=1}^{n} f(y_i, \theta) \quad \text{for } \theta \in \Omega$$

Example: Likelihood Function for Exponential Distribution. Since $f(x;\theta) = \frac{1}{\theta} e^{\frac{-y_i}{\theta}}$

$$L(\theta) = \prod_{i=1}^{n} \frac{1}{\theta} e^{\frac{-y_i}{\theta}} = \frac{1}{\theta^n} \exp\left(-\frac{1}{\theta}\sum_{i=1}^{n} y_i\right) = \theta^{-n} e^{\frac{-n\bar{y}}{\theta}} \quad \text{for } \theta > 0$$

$$l(\theta) = ln(L(\theta)) = -nln(\theta) + (\frac{-n\bar{y}}{\theta}) = -n(\log\theta + \frac{\bar{y}}{\theta})$$

$$\frac{dl(\theta)}{d\theta} = \frac{-n}{\theta} + \frac{n\bar{y}}{\theta^2} = \frac{n}{\theta^2}(\bar{y} - \theta) = 0 \implies \hat{\theta} = \bar{y} \text{ is the maximum likelihood estimate of } \theta$$

Example Likelihood Funciton for Gaussian Distribution.

$$L(\theta) = L(\mu, \sigma) = \prod_{i=1}^{n} f(y_i; \mu, \sigma) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\sigma} \exp\left[\frac{1}{-2\sigma^2}\sqrt{(y_i - \mu)}\right]$$
$$= (2\pi)^{\frac{-\pi}{2}} \sigma^{-n} \exp\left[\sum_{i=1}^{n} \frac{-1}{2\sigma^2}(y_i - \mu)^2\right]$$

Ignoring all the constant term without μ and θ.

$$L(\theta) = L(\mu, \sigma) = \sigma^{-n} \exp\left[\frac{-1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \mu)^2\right]$$

Since

$$\sum_{i=1}^{n}(y_i - \bar{y}) = \sum_{i=1}^{n}(y_i) - \sum_{i=1}^{n}(\bar{y}) = \sum_{i=1}^{n}(y_i) - ny_i = 0$$

and

$$\sum_{i=1}^{n}(y_i - \mu)^2 = \sum_{i=1}^{n}(y_i - \bar{y} + \bar{y} - \mu) \Rightarrow \sum_{i=1}^{n}(y_i - \bar{y})^2 - 2\sum_{i=1}^{n}(y_i - \bar{y})(\bar{y} - \mu)^{\ 2} - \sum_{i=1}^{n}(\bar{y} - \mu)^2 \Rightarrow \sum_{i=1}^{n}(y_i - \bar{y})^2 - \sum_{i=1}^{n}(\bar{y} - \mu)^2 = 0$$

$$L(\theta) = \sigma^{-n} \exp\left[\frac{-1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \bar{y})^2\right] \exp\left[\frac{-1}{2\sigma^2}\sum_{i=1}^{n}(\bar{y} - \mu)^2\right]$$

The log likelihood funciton is

$$l(\theta) = l(\mu, \sigma) = -n\log\sigma - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \bar{y})^2 - \frac{1}{2\sigma^2}n(\bar{y} - \mu)^2 \quad \text{for } \mu \in \mathbb{R} \text{ and for } \sigma > 0$$

Taks partial derivative to each variables

$$\frac{\partial(\theta)}{\partial\mu} = \frac{-1}{2\sigma^2}2n(\bar{y} - \mu) = \frac{-n}{\sigma^2}(\bar{y} - \mu) = 0 \Rightarrow \hat{\mu} = \bar{y}$$

Substitute $\hat{\mu} = \bar{y}$ into partial equation and complete expansion of $(y_i - \mu)^2$

$$\frac{\partial l(\theta)}{\partial \sigma} = \frac{-n}{\sigma} + \frac{1}{\sigma^3}\left(\sum_{i=1}^{n}(y_i-\bar{y})^2 - 2(y_i-\bar{y})(\bar{y}-\mu) + (\bar{y}-\mu)^2\right)$$
$$= \frac{-n}{\sigma} + \frac{1}{\sigma^3}(\sum_{i=1}^{n}(y_i-\bar{y})^2)$$

$$\frac{\partial l(\theta)}{\partial \sigma} = \frac{-n}{\sigma} + \frac{1}{\sigma^3}(\sum_{i=1}^{n}(y_i-\bar{y})^2) = 0$$

$$\frac{n}{\sigma} = \frac{1}{\sigma^3}\left(\sum_{i=1}^{n}(y_i-\bar{y})^2\right) \Rightarrow \frac{n}{\sigma}\sigma^3 = \sum_{i=1}^{n}(y_i-\bar{y})^2 \Rightarrow \sigma^2 = \frac{\sum_{i=1}^{n}(y_i-\bar{y})^2}{n} \Rightarrow \hat{\sigma} = \sqrt{\frac{\sum_{i=1}^{n}(y_i-y\bar{y})^2}{n}}$$

3.3 Likelihood Functions for Multinomial Models

Since the multinomial joint probability function is

$$f(y_1, y_2, \ldots, y_k; \theta) = \frac{n!}{y_1!y_2!\ldots y_k!}\prod_{i=1}^{k}\theta_i^{y_i} \quad \text{for } y_i = 0, 1, \ldots \text{where } \sum_{i=1}^{k} y_i = n$$

So the likelihood function for $\theta = (\theta_1, \ldots, \theta_k)$ based on data $y_1, \ldots, y_k$ is given by

$$L(\theta) = L(\theta_1, \ldots, \theta_k) = \frac{n!}{y_1!\ldots y_k!}\theta_i^{y_i}$$

Ignoring all the constant term withou θ.

$$L(\theta) = \prod_{i=1}^{k}\theta_i^{y_i}$$

The log likelihood function is

$$l(\theta) = \sum_{i=1}^{k} y_i \log\theta_i$$

Therefore the maximum likelihood estimation of $\theta_1, \ldots, \theta_k$ are

$$\hat{\theta} = \frac{y_i}{n}$$

Example: Suppose

$$P(Y_1 = y_1, Y_2 = y_2, Y_3 = y_3; \theta) = \frac{n!}{y_1!\ldots y_3!}y_1^{\theta_1}y_2^{\theta_2}y_3^{\theta_3}$$

For $\theta_1 = \alpha^2, \theta_2 = 2\alpha(1-\alpha), \theta_3 = 1-\alpha^2$ where $0 \le \alpha \le 1$. Then the likelihood function is

$$L(\alpha) = \frac{n!}{y_1!y_2!y_3!}(\alpha^2)^{y_1}(2\alpha(1-\alpha))^{y_2}\left[1-\alpha^2\right]^{y_3}$$

$$L(\alpha) == 2^{y_2}a^(2y_1 + y_2)(1-\alpha)^(y_2 + 2y_3)$$

The log likelihood function is

$$l(\alpha) = (2y_1 + y_2)\log\alpha + (y_2 + 2y_3)\log(1-\alpha) \Rightarrow \frac{dl(\alpha)}{d\alpha} = \frac{2y_1 + y_2}{\alpha} + \frac{y_2 + 2y_3}{1-\alpha} = 0$$

Then

$$\alpha = \frac{2y_1 + y_2}{2y_1 + y_2 + y_3} = \frac{2y_1 + y_2}{2n}$$

is the maximum likelihood of α.

Theorem: If $\bar{\theta} = \left(\bar{\theta}_1, \ldots, \bar{\theta}_k\right)$ is the maximum likelihood estimate of $\theta = (\theta_1, \ldots, \theta_k)$, then $g(\bar{\theta})$ is the maximum likelihood estimate of $g(\theta)$.

4 Estimation

4.1 Esitmator and Sampling Distribution

A estimator $\tilde{\theta}$ is a random variabel which is a function $\tilde{\theta} = g(Y_1, \ldots, Y_n)$ fo the random variables $Y_1, \ldots, Y_n$. The distribution of θ is called the sampling distirbution of the estimator.

Exmaple:Suppose to take a random sample $Y_1, \ldots, Y_n$ to estimate the unknown mean μ where $Y_i \sim G(\mu, \sigma), i = 1, \ldots, n$ and the maximum likelihood estimation of μ is

$$\tilde{\mu} = \bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i$$

$$\mathrm{Var}(\bar{Y}) = \mathrm{Var}(\frac{1}{n}\sum_{i=1}^{n} Y_i) = \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(Y_i) = \frac{1}{n^2}(n\sigma^2) = \frac{\sigma^2}{n} \Rightarrow sd(\bar{Y}) = \frac{\sigma}{\sqrt{n}}$$

then $\tilde{\mu} = \bar{Y} \sim G(\mu, \frac{\sigma}{\sqrt{n}})$.Dtermine how often of estimator $\tilde{\mu} = \bar{Y}$ within 0.01.

$$P(|\tilde{\mu} - \mu| \leq 0.01) = P(\mu{-}0.01 \leq \bar{Y} \leq \mu{+}0.01) = P\left(\frac{-0.01}{\frac{\sigma}{\sqrt{n}}} \leq \frac{\bar{Y} - \mu}{\frac{\sigma}{\sqrt{n}}} \leq \frac{0.01}{\frac{\sigma}{\sqrt{n}}}\right) \Rightarrow P\left(\mu - 0,01 \leq z \leq \mu + 0.01\right) \text{where } z \sim G(0,1)$$

Suppose $\sigma = 0.07$ and $n = 50$, then

$$P(|\tilde{\mu} - \mu| \leq 0.01) = P(\frac{-0.01\sqrt{50}}{0.07} \leq z \leq \frac{0.01\sqrt{50}}{0.07}) = P(-1.01 \leq z \leq 1.01) = 0.688$$

4.2 Confidence Intervals amd Pivotal Quantities

Example: Gaussian Distribution with unknown mean and known standard deviation Suppose that the observation$y_1, \ldots, y_n$ are random sample from $G(\mu, 2)$ distribution and $\mu = \mathbb{E}(Y_i)$ is unknown but $sd(Y_i) = 2$The maximum likelihood of μ is $\hat{\mu} = \bar{y}$ and the MLE of μ

$$\left[\bar{y} - 1.96(2)/\sqrt{n}, \bar{y} + 1.96(2/\sqrt{n})\right]$$

The interval is random interval which both endpoints are random variables. Since $Y \sim G(\mu, \frac{2}{\sqrt{n}})$

$$\begin{aligned} P\left(\mu \in \left[\bar{Y} - 1.96(2)/\sqrt{n}, \bar{Y} + 1.96(2)/\sqrt{n}\right]\right) &= P(\bar{Y} - 1.96(2)/\sqrt{n} \leq \mu \bar{Y} + 1.96(2)/\sqrt{n}) \\ &= P(-1.96 \leq \frac{\bar{y} - \mu}{2/\sqrt{n}} \leq 1.96) \\ &= P(-1.96 \leq Z \leq 1.96) \text{where } Z \in G(0,1) \\ &= 0.95 \end{aligned}$$

Interval Estimator: The interval estimator $[L(Y), U(Y)]$ has property that

$$P(\theta \in [L(Y), U(Y)]) = P(L(Y) \leq \theta U(Y)) = p$$

The interval estimate $[L(y), U(y)]$ $a100p\%$ confidence interval for θ and p is condifence coefficient

Pivotal Quantity: $Q = Q(Y; \theta)$ is a funciton of data Y and the unknown parameter θ such that the distirbution of the random variable Q is fully known. The probability statements such as $P(Q \leq b)$ and $P(Q \leq a)$depend on a and b but not on θ or any unknown information.

4.2.1 Asymptotic Gaussian Pivotal Quantities

Suppose $\bar{\theta}$ is point estimator of the unknown parameter of θ. Supppose that the Central Limit Theorem can be used to obtain the result that

$$\frac{\bar{\theta}-\theta}{g(\theta)-\sqrt{n}}$$

has approximately $G(0,1)$ distribution for large n where $\mathbb{E}(\bar{\theta})=\theta$ and $sd(\bar{\theta})=h(\theta)/\sqrt{n}$ for some real valued $g(\theta)$.We replace θ by $\bar{theta}$ in the denominator then

$$Q_n(\bar{\theta},\theta)=\frac{\bar{\theta}-\theta}{g(\bar{\theta})/\sqrt{n}}$$

alse has approximately a $G(0,1)$ distribution for large n. Therefore $Q(\bar{\theta},\theta)$ is an asymptotic Gaussian pivotal quantity which can be used to construct approximate confidence interval for θ.

4.3 The chi-squared and t Distributions

4.3.1 Gamma Function

$$\Gamma(\alpha)=\int_0^\infty y^{\alpha-1}e^{-y}\,\mathrm{d}y$$

- $\Gamma(\alpha)=(\alpha-1)\Gamma(\alpha-1)$
- $\Gamma(\alpha)=(\alpha-1)!,\text{for }\alpha\in\mathbb{N},a>0$
- $\Gamma(\frac{1}{2})=\sqrt{n}$

The $\chi^2(k)$ distribution is a continuous family of distribution $(0,\infty)$ with probability density funciton of the form

$$f(x;k)=\frac{1}{2^{\frac{k}{2}}\Gamma(\frac{k}{2})}x^{\frac{k}{2}-1}e^{\frac{-x}{2}}\qquad\text{for } x>0,\ \text{where } k\in\{k=1,2,\dots\}$$

Denote $X\sim\chi(k)^2$

- for $k=2$ substitute into equation $f(x;2)=\frac{1}{2\Gamma(1)}x^0e^{\frac{-x}{2}}=\frac{1}{2}e^{\frac{-x}{2}}$ equivalent to $Exponential(2)$
- for $k>2$ then *pdf* is unimodal with maximum value at $x=k-2$.
- for $k\geq 30$ then *pdf* resembles that of $N(k,2k)pdf$

If $X\sim\chi^2$, Then

$$\mathbb{E}(X)=k\text{ and }\mathrm{Var}(X)=2k$$

$$\begin{aligned}
\mathbb{E}(X^j)&=\int_0^\infty x^j\frac{1}{2^{\frac{k}{2}}\Gamma(\frac{k}{2})}x^{(\frac{k}{2}-1)}e^{\frac{k}{2}}\,\mathrm{d}x\\
&=\frac{1}{\Gamma(\frac{k}{2})}\int_0^\infty\frac{1}{2^{\frac{k}{2}}}x^{(\frac{k}{2}+j-1)}e^{\frac{k}{2}}\,\mathrm{d}x\qquad\text{Let } y=\frac{x}{2}\Rightarrow dx=2dy\\
&=\frac{1}{\Gamma(\frac{k}{2})}\int_0^\infty 2^{-\frac{k}{2}}(2y)^{\frac{k}{2}+j-1}2\,\mathrm{d}y\\
&=\frac{2^j}{\Gamma(\frac{k}{2})}\int_0^\infty y^{(}\frac{k}{2}+j-1)\,\mathrm{d}y\\
&=\frac{2^j\Gamma(\frac{k}{2}+j)}{\Gamma(\frac{k}{2})}
\end{aligned}$$

For $j=1$, then

$$\mathbb{E}(X)=\frac{2\Gamma(\frac{k}{2}+1)}{\Gamma\frac{k}{2}}=\frac{2\frac{k}{2}\Gamma(\frac{k}{2})}{\Gamma(\frac{k}{2})}=k$$

$$\mathbb{E}(X^2) = \frac{2^2\Gamma(\frac{k}{2}+2)}{\Gamma(\frac{k}{2})} = \frac{4\Gamma((\frac{k}{2}+1)+1)}{\Gamma(\frac{k}{2})} = 4(\frac{k}{2}+1)(\frac{k}{2}) = k(k+1)$$

$$\begin{aligned}\int_0^\infty \frac{1}{2^{\frac{k}{2}}\Gamma(\frac{k}{2})} x^{\frac{k}{2}-1}e^{\frac{-x}{2}}\,\mathrm{d}x &= \int_0^\infty \frac{1}{2^{\frac{k}{2}}\Gamma(\frac{k}{2})}(2y)^{\frac{k}{2}-1}e^{-y}2y\,\mathrm{d}y \\ &= \frac{2^{\frac{k}{2}-1+1}}{2^{\frac{k}{2}}\Gamma(\frac{k}{2})}\int_0^\infty y^{\frac{k}{2}-1}e^{\frac{-y}{2}}\,\mathrm{d}y \\ &= \frac{\Gamma(\frac{k}{2})}{\Gamma(\frac{k}{2})} \\ &= 1\end{aligned}$$

mgf of χ^2 distirbution:

$$\begin{aligned}M(t) = \mathbb{E}(e^{tX}) &= \int_0^\infty e^{tx}f(x,k)\,\mathrm{d}x \\ &= \int_0^\infty e^{tx}\frac{1}{2^{\frac{k}{2}}\Gamma(\frac{k}{2})}x^{(\frac{k}{2}-1)}e^{\frac{-x}{2}}\,\mathrm{d}x \\ &= \int_0^\infty \frac{1}{2^{\frac{k}{2}}\Gamma(\frac{k}{2})}x^{(\frac{k}{2}-1)}e^{-x(1-2t)}\,\mathrm{d}x \qquad \text{let } u = x\frac{1-2t}{2} \text{ and } \frac{2u}{1-2t} \Rightarrow dx = \frac{2du}{1-2t} \\ &= \int_0^\infty \frac{1}{2^{\frac{k}{2}}\Gamma(\frac{k}{2})}\left(\frac{2u}{1-2t}\right)^{\frac{k}{2}-1}e^{-u}\,\mathrm{d}u \\ &= \int_0^\infty \frac{1}{2^{\frac{k}{2}}\Gamma(\frac{k}{2})}\left(\frac{2}{1-2t}\right)^{\frac{k}{2}}u^{\frac{k}{2}-1}e^{-u}\,\mathrm{d}u \\ &= \frac{1}{\Gamma(\frac{k}{2})(1-2t)^{\frac{k}{2}}}\int_0^\infty u^{\frac{k}{2}-1}e^{-u}\,\mathrm{d}u \\ &= \frac{\Gamma(\frac{k}{2})}{\Gamma(\frac{k}{2})(1-2t)^{\frac{k}{2}}} \\ &= (1-2t)^{-\frac{k}{2}} \qquad \text{for } t < \frac{1}{2}\end{aligned}$$

Theorem: The additive of χ^2 variables.
Let $W_1, \ldots, W_n$ be independent random variables with $W_i \sim \chi^2(k_i)$. Then

$$S = \sum_{i=1}^n W_i \sim \chi^2\left(\sum_{i=1}^n k_i\right)$$

Proof. Let $S = \sum_{i=1}^n W_i$. Since$W_1, \ldots, W_n$ are independent random variables with $W_i \sim \chi^2(k_i)$, then the mgf of S

$$\begin{aligned}M_S(t) &= \prod_{i=1}^n M_{W_i}(t) \\ &= \prod_{i=1}^n (1-2t)^{\frac{-k_i}{2}} \\ &= (1-2t)^{\left(\sum_{i=1}^n k_i\right)} \\ &= (1-2t)^{\frac{-1}{2}\sum_{i=1}^n k_i}\end{aligned}$$

Thus $S = \sum_{i=1}^n W_i \sim \chi^2(k_i)$ □

Theorem: If $Z \sim G(0,1)$ then the distribution of $W = Z^2$ is $\chi^2(1)$

Proof. Suppose that $W = Z^2$ where $Z \sim G(0,1)$. Let Φ represent the *cdf* of $G(0,1)$random variables and let ϕ represent the *pdf* of $G(0,1)$ of random variable. For $w > 0$

$$P(W \geq w) = P(-\sqrt{w} \leq z \leq \sqrt{w}) = \Phi(\sqrt{w}) - \Phi(-\sqrt{w})$$

Then the *pdf* of W is

$$\begin{aligned}
\frac{d}{dw}P(W \leq w) &= \frac{d}{dw}\left[\Phi(\sqrt{w}) - \Phi(-\sqrt{w})\right] \\
&= \left[\phi(\sqrt{w}) - \phi(-\sqrt{w})\right]\left(\frac{1}{2}w^{\frac{-1}{2}}\right) \\
&= 2\left[\frac{1}{\sqrt{2\pi}}e^{\frac{-(\sqrt{w}^2)}{2}}\frac{1}{2\sqrt{w}}\right] \\
&= \left[\frac{1}{\sqrt{2\pi}}e^{\frac{-w}{2}}w^{\frac{-1}{2}}\right] \\
&= \frac{1}{2^{\frac{1}{2}}\Gamma(\frac{1}{2})}w^{\frac{1}{2}-1}e^{\frac{-w}{2}}
\end{aligned}$$

Thus the probability density function of a $\chi^2(1)$ random variable as required

□

Corollary: If $Z_1, \ldots, Z_n$ are mutually independent $G(0,1)$ random variables and $S = \sum_{i=1}^{n} {Z_i}^2$, then $S \sim \chi^2(n)$

Proof. Since $Z_i \sim G(0,1)$ then by previous theorem $W = {Z_i}^2 \sim \chi^2(1)$. According to the additive of χ^2 variables, $W_1, \ldots, W_n$ are mutually independent random variable, let $S = \sum_{i=1}^{n} W_i$. Therefore

$$S = \sum_{i=1}^{n} W_i \sim \chi^2\left(\sum_{i=1}^{n} 1\right) = \chi^2(n)$$

□

Theroem: If $X \sim \chi^2(n_1), Y \sim \chi^2(n_2)$ also X and Y are mutually independent. Then

$$X + Y \sim \chi^2(n_1 + n_2)$$

4.3.2 Student's t Distribution

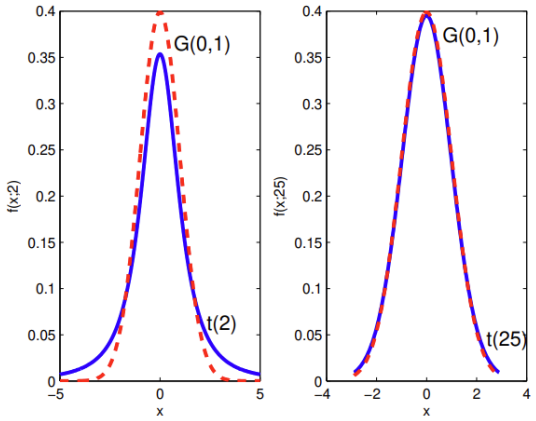
The *pdf* of Student's t distribution:

$$f(t;k) = c_k\left(1 + \frac{t^2}{k}\right)^{-\frac{k+1}{2}} \qquad \text{for } t \in \mathbb{R} \text{ and } k = 1, 2, \ldots$$

where

$$c_k = \frac{\Gamma(\frac{k+1}{2})}{\sqrt{k\pi}\Gamma(\frac{k}{2})}$$

The parameter k is the degrees of freedom. Denote as $T \sim t(k)$ to indicate that the random variable T has t distribution with k degrees of freedom.For smaller k(such as $k = 2$), the peak of t distribution is lower and has "fatter tails" at both ends. Implies that the probability of extreme value is higher than standard normal distribution. as $t- > \infty, f(t,k)- > \frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}$



Theroem:Suppose $Z \sim G(0,1)$ and $U \sim \chi^2(k)$ independently. Let

$$T = \frac{Z}{\sqrt{\frac{U}{k}}}$$

Then T has a Student's t distribution with k degrees of freedom.

4.4 Likelihood-Based Confidence Intervals

The relative likelihood

$$R(\theta) = \frac{L(\theta)}{L(\hat{\theta})} \qquad \text{for } \theta \in \Omega$$

is the function of the maximum lieklihood estimate $\hat{\theta}$. Use random variable $\bar{\theta}$ to substitute $\hat{\theta}$ and define $\lambda(\theta)$

$$\lambda(\theta) = -2\log\left[\frac{L(\theta)}{L(\bar{\theta})}\right]$$

where $\bar{\theta}$ is the maximum likelihood estimator and $\lambda(\theta)$ is the likelihood ratioi statistic.

Theorem: If $L(\theta)$ is based on $Y = (Y_1, \ldots, Y_n)$ a ranodm sample of size n and if θ is the true value of the each parameter, then thedsitribution of $\lambda(\theta)$ converges to $\chi^2(1)$ distribution as $n- > \infty$.

Theorem: A $100p\%$ lieklihood interval is an approximate $100q\%$ confidence interval where $q = wP(Z \geq \sqrt{-2logp} - 1)$ and $Z \sim N(0,1)$.

Proof. A $100p\%$ likelihood interval is defined by $\{\theta \mid R(\theta) \geq p\}$, which can be rewritten as:

$$\{\theta \mid R(\theta) \geq p\} = \left\{\theta \mid -2\log\left[\frac{L(\theta)}{L(\hat{\theta})}\right] \leq -2\log p\right\}$$

$$\begin{aligned}
P(\Lambda(\theta) \leq -2\log p) &= P\left(-2\log\left[\frac{L(\theta)}{L(\hat{\theta})}\right] \leq -2\log p\right) \\
&= P(W \leq -2\log p) \qquad \text{for } w \sim \chi^2(1) \text{ from previous theorem} \\
&= P(-\sqrt{-2\log p} \leq Z \leq \sqrt{-2\log p}) \qquad \text{since } Z \sim N(0,1) \\
&= P(Z \leq \sqrt{-2\log p}) - \left[1 - P(-\sqrt{-2\log p} \leq Z)\right] \\
&= 2P(Z \leq \sqrt{-2\log p}) - 1
\end{aligned}$$

□

Example: If $p = 0.1$, then

$$\begin{aligned} q &= 2P\left(\sqrt{-2\log(0.1)}\right) - 1 \qquad \text{where } Z \sim N(0,1) \\ &= 2P(Z \leq 2.15) - 1 \\ &= 0.96844 \end{aligned}$$

4.5 Condifence Intervals for Parameters in the $G(\mu, \sigma)Model$